RWorksheet_Piano#3a

```r
#1a
first_eleven <-LETTERS[1:11]
```

```r
#1b
oddletters <-LETTERS[seq(1,26, by = 2)]
oddletters
```

```
##  [1] "A" "C" "E" "G" "I" "K" "M" "O" "Q" "S" "U" "W" "Y"
```

```r
#1c
vowels <-LETTERS[LETTERS %in% c (1,5,9,15,21)]
```

```r
#1d
last5 <-LETTERS[22:25]
```

```r
#1e
between <- LETTERS[15:24]
```

```r
#2a
city <-c("Tuguegarao City", "Manila", "Iloilo City", "Tacloban", "Samal Island", "Davao City")
temp <- c(42,39,34,34,30,27)
```

```r
#2b
temp <- c(42,39,34,34,30,27)
```

```r
#3c
city_temp <- data.frame(city, temp)
```

```r
#3d
names(city_temp)[1]<-"City"
names(city_temp)[2]<-"Temperature"
city_temp
```

```
##              City Temperature
## 1 Tuguegarao City          42
## 2          Manila          39
## 3     Iloilo City          34
## 4        Tacloban          34
## 5    Samal Island          30
## 6      Davao City          27
```

```r
#3e
str(city_temp)
```

```
## 'data.frame':    6 obs. of  2 variables:
##  $ City       : chr  "Tuguegarao City" "Manila" "Iloilo City" "Tacloban" ...
##  $ Temperature: num  42 39 34 34 30 27
```

```r
#The output shows data frame containing six cities with their corresponding temperature
```

```r
#3f
#The content of 3 and 4  is Iloilo and Tacloban and they have the same temperature
```

```r
#3g
print(city_temp[1,])
```

```
##                City Temperature
## 1 Tuguegarao City          42
```

```r
print(city_temp[6,])
```

```
##          City Temperature
## 6 Davao City          27
```

#MATRICES

```r
#1
matrix(c(5,6,7,4,3,2,1,2,3,7,8,9),nrow =2)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]    5    7    3    1    3    8
## [2,]    6    4    2    2    7    9
```

```r
matrix(data=c(3,4,5,6,7,8),3,2)
```

```
##      [,1] [,2]
## [1,]    3    6
## [2,]    4    7
## [3,]    5    8
```

```r
diag(1,nrow = 6,ncol = 5)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    0    0    0    0
## [2,]    0    1    0    0    0
## [3,]    0    0    1    0    0
## [4,]    0    0    0    1    0
## [5,]    0    0    0    0    1
## [6,]    0    0    0    0    0
```

```r
diag(6)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]    1    0    0    0    0    0
## [2,]    0    1    0    0    0    0
## [3,]    0    0    1    0    0    0
## [4,]    0    0    0    1    0    0
## [5,]    0    0    0    0    1    0
## [6,]    0    0    0    0    0    1
```

```r
#2a
one.eight <- matrix(c(1:8, 11:14),3,4)
one.eight
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    4    7   12
## [2,]    2    5    8   13
## [3,]    3    6   11   14
```

```r
#2b
one.eight2 <- one.eight * 2
one.eight2
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    2    8   14   24
## [2,]    4   10   16   26
## [3,]    6   12   22   28
```

```r
#2c
one.eight[2,]
```

```
## [1]  2  5  8 13
```

```r
one.eight[2,]
```

```
## [1]  2  5  8 13
```

```r
#2d
one.eight[1:2, 3:4]
```

```
##      [,1] [,2]
## [1,]    7   12
## [2,]    8   13
```

```r
#2e
one.eight[3, 2:3]
```

```
## [1]  6 11
```

```r
#2f
one.eight[, 4]
```

```
## [1] 12 13 14
```

```r
#2g
dimnames(one.eight2) <-list(c("isa", "dalawa", "tatlo"), c("uno", "dos", "tres", "quatro"))
one.eight2
```

```
##        uno dos tres quatro
## isa      2   8   14     24
## dalawa   4  10   16     26
## tatlo    6  12   22     28
```

```r
#3a
arrayval <-c(1,2,3,4,5,6,7,8,9,0,3,4,5,1)
arraydata <-array(rep(arrayval, 2), dim=c(2,4,3))
arraydata
```

```
## , , 1
##
##      [,1] [,2] [,3] [,4]
## [1,]    1    3    5    7
## [2,]    2    4    6    8
##
## , , 2
##
##      [,1] [,2] [,3] [,4]
## [1,]    9    3    5    1
## [2,]    0    4    1    2
##
## , , 3
##
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    3    5    7    9
## [2,]    4    6    8    0
```

```r
#
```

```r
#3b
dim(arraydata)
```

```
## [1] 2 4 3
```

```r
#3c
rownames(arraydata)<-c("a", "b")
colnames(arraydata)<-c("A", "B", "C", "D")
```